Journal of Pediatric Nursing 65 (2022) 116–123

Contents lists available at ScienceDirect

Journal of Pediatric Nursing

journal homepage: www.pediatricnursing.org

Moving up: Healthcare transition experiences of adolescents and young adults with cystic fibrosis

Katherine South, BSN, RN, CPN [a,*], Maureen George, PhD, RN, AE-C, FAAN [a], Hossein Sadeghi, MD [b], Victoria Piane, MSN, CPNP [b], Arlene Smaldone, PhD, CPNP, CDCES, FAAN [a]

[a] Columbia University School of Nursing, 560 West 168th Street, New York, NY 10032, United States of America
[b] Columbia University Irving Medical Center, Division of Pediatric Pulmonology and Sleep Medicine, 3959 Broadway, New York, NY 10032, United States of America

ARTICLE INFO

Article history:
Received 15 November 2021
Revised 18 March 2022
Accepted 20 March 2022

Keywords:
Healthcare transition
Cystic fibrosis
Qualitative
Chronic illness
Adolescent

ABSTRACT

Purpose: The experience of healthcare transition from pediatric to adult care in cystic fibrosis (CF) remains poorly understood, particularly among racially and ethnically diverse adolescents and young adults (AYAs) with CF. The objective of this qualitative study was to explore the perspectives of a diverse sample of AYAs with CF at one urban academic medical center regarding healthcare transition.

Design and methods: Guided by qualitative descriptive methodology, we purposively selected AYAs who represented the pre and post transition experience: some AYAs had experienced the transition preparation program CF R.I.S.E. Demographic information and responsibility for self-management behaviors were collected using an online survey. Semi-structured video interviews were conducted following an iterative interview guide. A codebook directed inductive coding. QSR NVivo Version 12 software was used to organize the data.

Results: 12 AYAs with CF were enrolled (25% female, 25% Black AYA, 33% Hispanic/Latina/o AYA, 50% White AYA; mean age 20.8 years). Three themes were identified: *independent care of the whole self*, *preparing for change and the unknown* and *transition experiences vary*.

Conclusions: Not all participants experienced a smooth transition. Participants identified suggestions for the development of transition preparation interventions, specifically around involving AYAs in transition decisions and beginning transition preparation early in adolescence.

Practice implications: Participants expressed uncertainty about transition when they felt little control over the process or lacked sufficient information about adult care. Therefore, comprehensive early transition preparation for all AYAs with CF with a focus on involving AYAs in transition decisions is recommended.

© 2022 Elsevier Inc. All rights reserved.

Introduction

Healthcare transition, or the planned and purposeful movement from pediatric to adult care, is an important life event for adolescents and young adults (AYAs) with cystic fibrosis (CF) (Rosen et al., 2003). Once a disease of childhood, people with CF now live longer, healthier lives; today, more than half of the CF population in the United States is over 18 years old (Cystic Fibrosis Foundation, 2019; Goralski et al., 2017). The proportion of young adults with CF is expected to continue to increase, aided by the increased availability of highly effective cystic fibrosis transmembrane conductance regulator (CFTR) modulator therapies that target specific genetic mutations (Middleton & Taylor-Cousar, 2021). While CF is more prevalent among White individuals, Black and Latina/o individuals are also affected by the disease and have increased morbidity and mortality due, in part, to more frequently having different genetic mutations from the ones that current CFTR therapies target (McGarry & McColley, 2021). Other factors including socioeconomic status and implicit bias are likely to also contribute to these disparities (McGarry et al., 2019).

In order to receive developmentally appropriate care, this growing population of AYAs with CF must transition from pediatric to adult care, typically around 18–21 years of age (Goralski et al., 2017). Distinct from transfer which is the one-time act of moving from pediatric to adult care, a successful healthcare transition is a process that promotes independent self-management and wellness across the lifespan, allows AYAs uninterrupted access to medical care, and prepares AYAs for an adult model of care (Davidson & Doyle, 2021; Lanzkron et al., 2018; Rosen et al., 2003; Society for Adolescent Health and Medicine, 2020; White & Cooley, 2018). However, advances in healthcare transition have not kept up with medical advances which have prolonged the lifespan of young people with chronic conditions such as CF (Betz et al., 2021). AYAs with CF who do not experience a smooth and

* Corresponding author.
E-mail address: kts2133@cumc.columbia.edu (K. South).

https://doi.org/10.1016/j.pedn.2022.03.007
0882-5963/© 2022 Elsevier Inc. All rights reserved.

prepared transition to adult care are at increased risk of gaps in continuity of care and clinical deterioration following transfer to adult care (Sawicki et al., 2018). For example, a retrospective cohort study of 1946 AYAs with CF found that 13% experienced a gap greater than 365 days between their last pediatric visit and their first visit at the adult CF clinic (Sawicki et al., 2018). This is a major deviation from the recommended quarterly clinic visits for this population and is associated with poorer lung function compared to those who maintain continuity of care (Johnson et al., 2003; Padman et al., 2007; Sawicki et al., 2018). Therefore, supporting AYAs with CF and their families and preparing them for transition is key to preventing adverse outcomes following transfer to adult care.

Equipping AYAs with self-management skills to promote optimal transition outcomes is the primary goal of transition preparation programs (Coyne et al., 2017). To that end, CF transition programs should begin the preparation process in early adolescence, assessing AYA transition readiness, ensuring that the AYA has an understanding of CF and its treatments, assessing and addressing issues of adherence, addressing mental health concerns and ensuring that the final transfer of care from the pediatric to the adult care setting is coordinated and completed (Okumura & Kleinhenz, 2016). CF R.I.S.E. (Responsibility. Independence. Self-Care. Education) is a transition preparation program which meets the Cystic Fibrosis Foundation's education committee's guidelines and standards for educational programs (Cystic Fibrosis Education & Support Resource: CF R.I.S.E., 2021). At this center, CF R.I.S.E. is a nurse-led intervention which uses web-based knowledge assessments for AYAs and responsibility checklists focusing on shared vs. independent CF self-management tasks for AYAs and their caregivers (Baker et al., 2015; Cystic Fibrosis Education & Support Resource: CF R.I.S.E., 2021). To our knowledge, only a single study (Baker et al., 2015) has explored perceptions of the CF R.I.S.E. program; however, AYA perspectives were not included. How AYAs with CF view programs such as CF R.I.S.E. and their experience of the healthcare transition preparation process is poorly understood, particularly among AYAs who may benefit from longer survival due to CFTR therapy and among those who largely will not, such as racially- and ethnically diverse AYAs with CF (Al-Yateem, 2012; Brumfield & Lansbury, 2004). To close this gap in our understanding of transition perspectives of AYAs with CF, this study explored the experience of healthcare transition preparation and transfer of care for AYAs with CF from one urban academic medical center.

Methods

Guiding qualitative methodology

This study was guided by qualitative descriptive methodology. With origins in naturalistic inquiry, this methodology is appropriate when the researcher aims to gain an insider's perspective of a phenomenon that is poorly understood, (Sandelowski, 2000), such as healthcare transition experiences among diverse AYAs with CF. Unlike more interpretive methodologies, qualitative descriptive focuses on presenting facts and describing events using simple and accessible language (Bradshaw et al., 2017; Sandelowski, 2000; Sullivan-Bolyai et al., 2005), privileging the voice of the participant with minimal interpretation from the researcher (Sandelowski, 2000). In accordance with this methodology, data collection and analysis occur simultaneously with the researchers remaining 'close to the data' in order to credibly render a description of the phenomenon (Sandelowski, 2000). Qualitative descriptive approaches are commonly used in health research (Bradshaw et al., 2017; Sullivan-Bolyai et al., 2005).

Participants

Purposive sampling techniques were used to select participants who could provide rich information about how the phenomenon of transition is experienced (Patton, 1990; Sandelowski, 2000). All eligible participants in either the pediatric or adult outpatient CF centers at one urban, academic medical center were invited to enroll. In 2019, 9.6% of the center's pediatric patients identified as Black and 36.8% as Hispanic or Latina/o. At the adult center, 3.5% of patients identify as Black and 7.6% as Hispanic or Latina/o.

Eligibility criteria stipulated that study participants needed to: 1) have either begun transition preparation, with a planned transfer to CF adult care in the next 3 years or have completed their transfer to adult care in the past 5 years; 2) be 15–24 years of age; 3) be English speaking; and 4) have the cognitive/verbal capacity to participate in the interview.

Recruitment

Eligible AYAs were mailed informational letters introducing the study on behalf of their CF physician. If an AYA did not opt-out of the study within 2 weeks of receiving the letter, a member of the research team called participants to introduce the study. If the AYA was a minor, the parent was the first point of contact, using the same opt-out letter approach. Efforts during the recruitment process were made to sample across both pre- and post-transfer experiences.

Data collection

Self-report of demographic information and self-management responsibility was collected using an online survey. Then semi-structured interviews were conducted using Zoom Pro, a HIPAA- compliant videoconferencing platform. An iterative interview guide (Table 1) directed the interviews (Kallio et al., 2016) and was pilot tested prior to use. Interviews were conducted by one researcher, a pediatric nurse with extensive CF experience who was unknown to the participants.

The interviews were audio and video recorded. Video recordings were reviewed immediately following the interviews to create field notes of non-verbal behaviors and settings. Video files were deleted within one hour of the interview, following the creation of field notes. The audio files were electronically transferred for professional transcription. These transcripts were then cleaned and stripped of any personal identifying information. After a clean transcript was produced, the audio files were securely stored for one year.

Ethical considerations

Informed consent was obtained after the researcher explained the study and answered participant questions. The consent process was conducted remotely over Zoom Pro, a HIPAA compliant, secure videoconferencing platform, and written consent was documented electronically using e-signatures. In the case of minors, informed consent was obtained from a parent prior to contacting the participant, and youth assent was obtained prior to the interview. This study was approved by

Table 1
Interview Guide.

1) Moving up from the pediatric to the adult CF clinic is a big milestone for people with CF, tell me about your upcoming/recent move to the adult CF clinic.
2) What do you think is the right age for the first visit at the adult CF clinic?
3) Over what period of time do you think preparing to move up to the adult clinic should occur?
4) Thinking about your own experience preparing to move up to the adult clinic, what should be done differently?
5) We know that CF teams often prescribe hours of treatments each day which can make it hard to do everything every day. How do you prioritize your care?
6) Part of moving to adult care is the expectation that you assume responsibility for your health. Tell me what it is like to assume this responsibility.
7) When it comes to taking care of yourself, what help do you need?
8) Is there anything else you'd like to share that I haven't asked about already?

the university's Institutional Review Board. Participants received a $50 gift card upon completion of the study.

Data analysis

Data collection ended when we had achieved data saturation, a measure of data adequacy that indicates that what can be known about the phenomenon has been learned, signaling an end to recruitment. The saturation table (Table 2) was created concurrent with ongoing data analysis (Kerr et al., 2010) and confirmed that data saturation had been reached after nine interviews. Three additional interviews were completed as additional assurance that data saturation had been reached. Saturation likely occurred with our small number of participants because of the homogeneity of our sample: participants had CF and transition experiences in common (Guest et al., 2006).

QSR NVivo Version 12 software was used to manage the data. Inductive coding was performed using conventional content analysis (Hsieh & Shannon, 2005) guided by an iterative codebook (MacQueen et al., 1998). While the Expanded Socioecological model for AYA Readiness for Transition (Expanded SMART) (Pierce et al., 2017) guided the general development of the study, we purposefully bracketed this conceptual framework so as to prevent biasing our findings and arriving at premature data analysis. Codes were assigned to the data as they emerged from reading the interview transcripts (Elo & Kyngäs, 2008). Meaning units were organized into codes, then shared codes were collapsed into categories; themes and subthemes were then formed from shared categories (Bradshaw et al., 2017). To develop a codebook, three authors independently coded the first two interviews then created definitions and boundaries (when to use; when not to use) for the codes (MacQueen et al., 1998). This codebook guided the first author in coding the remaining interviews. The research team met weekly to discuss the codes and to audit the coding process. Coding conflicts were resolved through these peer debriefing meetings. When describing the frequency of the findings and themes, the following categories are used: most (at least 75% of participants), many (>50% of participants), some (> 25% of participants), few (<25% of participants). As consistent with the philosophical underpinnings of the qualitative paradigm, themes reported with greater frequency were not deemed more important.

Trustworthiness

To ensure rigor, member checking was conducted during data collection (Birt et al., 2016; Guba, 1981). To ensure that emerging themes resonated with participants, preliminary findings were shared in remaining interviews and participant feedback was elicited. In addition, reflexivity was practiced throughout the data collection and analysis process in order to reduce bias stemming from the researcher's previous clinical nursing experiences working with AYAs with CF and to decrease the researcher's influence on the research process (Ahern, 1999).

Results

Interviews were conducted over a 5-month period and each interview lasted an average of 36 min (range 27–50 min). Of 21 eligible AYAs, 12 (25% female; 50% White, 25% Black, 33% Hispanic or Latina/o) participated in the study. Participant ages ranged from 15 to 24 years with a mean age of 20.8 years. Half had completed their transition and were receiving care at the adult CF center. Table 3 presents additional demographic characteristics of the sample. Of the AYAs who declined participation ($n = 5$) or were unable to be contacted ($n = 4$), 44% had completed their transition and 44% were female.

We also collected information on CF self-management responsibility by asking what percentage of time participants assumed self-management responsibility for different tasks. Self-management responsibility varied by transfer status, with post-transfer participants

Table 2
Saturation Table.

Theme	Category	Code	Interview Number											
			1	2	3	4	5	6	7	8	9	10	11	12
Independent Care of the Whole Self	Self-Management	Lifestyle Integration	X											
		Rationalization	X											
		Moving Towards Independence	X											
		"You know what you got to do"		X										
		Treatment Routine	X											
		Barriers and Facilitators to Care	X											
		Feeling Empowered	X											
		"It doesn't matter what color you are"								X				
	Mind and Body	Mind and Body	X											
	Disclosure & Peers	Disclosure	X											
		"I surround myself with the right people"			X									
Preparing for Change and the Unknown	The Language of Moving Up	The Language of Moving Up		X										
	CF RISE	CF RISE				X								
	Getting Ready	Fear of the Unknown	X											
		The Right Time for Me	X											
		Family Involvement	X											
		Transition Preparation Suggestions		X										
		I'm Not a Child Anymore						X						
		Barriers to Transition			X									
		Power Sharing		X										
		Ready for a New Provider							X					
		Redefining Roles			X									
		Shifting Responsibility	X											
		Transition Preparation		X										
		Transition Positives									X			
Transition Experiences Vary	Moving Up	Seamless Transfer	X											
		Sudden Transfer					X							
		Transfer as Abandonment								X				
		Building Trust		X										
		Enhanced Communication	X											
		Similarities & Differences		X										

Note: Each X denotes that a code contributing to a category was first noted in the corresponding interview

Table 3
Participant Demographics ($N = 12$).

	N	%	Mean	SD
Age (mean ± standard deviation)			20.8	2.8
Pre-transition	6	50%		
Post-transition	6	50%		
Female	3	25%		
Black	3	25%		
Declined to State	2	16.7%		
Native Hawaiian/ Pacific Islander	1	8.3%		
White	6	50%		
Hispanic/Latina/o Ethnicity	4	33%		
Working/Volunteering	4	33%		
Attending School	6	50%		
High School Diploma or General Educational Diploma	11	91.6%		
Lives with Parent(s)	8	66.7%		
Attends CF Clinic Alone	6	50%		
Percent of time AYA Takes Responsibility for Self-Management Tasks:	Pre-transfer	Post-transfer		
Cleaning Treatment Equipment	74%	86%		
Scheduling CF Clinic Appointments	42%	100%		
Calling in Prescription Refills	45%	88%		

reporting greater self-management responsibility. On average, pre-transfer participants reported being responsible for cleaning their treatment equipment (i.e.: nebulizer) 74% of the time, calling their CF provider for prescription refills 45% of the time and scheduling CF center appointments 42% of the time. Post-transfer participants reported being responsible for cleaning their treatment equipment 86% of the time, calling their CF provider for prescription refills 88% of the time and scheduling CF center appointments 100% of the time. All pre-transfer and most post-transfer participants reported sharing CF management responsibilities with a parent if they had not already assumed full independent responsibility for tasks themselves.

Themes

Three themes were identified from our inductive analysis: *independent care of the whole self* (importance of assuming self-management responsibility prior to transfer)*; preparing for change and the unknown* (the process of transition preparation and coping with the associated change and uncertainty); and *transition experiences vary* (spectrum of transition experiences from planned to abrupt).

Independent care of the whole self

Participants described the triumphs and challenges that they encountered as they took on increased self-management responsibility throughout their transition process. For one participant, self-management triumphs included becoming more confident with everyday responsibilities related to their CF:

> Once you start doing that, you feel more independent and a little bit more confident going into that stuff because you're like, okay, I'm doing it on my own. I don't need my parents or my grandparents anymore, and you feel a little bit more confident doing it.
> (#2, non-Hispanic/Latina/o, White AYA, post-transfer)

For another participant, successful independent self-management came with feelings of empowerment and accomplishment: “Just ordering my medicine makes me feel like happy like I achieved something.” (#5, non-Hispanic/Latina/o, Black AYA, post-transfer) However, becoming more independent was also noted to be a gradual and sometimes challenging process as a participant who had recently transferred to adult care shared: “Sometimes, I don't feel great about it. Sometimes, I wish I were more independent, but it's also, I guess, it's a gradual process, and I've got to be okay with it.” (#7, non-Hispanic/Latina/o, White AYA, post-transfer)

For all participants, a major theme surrounding becoming independent with CF self-management was the shift of responsibility from parent to child. One participant described how shifting responsibility early on was helpful:

> It started out, when I was like, I think I was about 13 or 14, my mom would sit next to me, and I would call in my prescriptions to the local pharmacy. So, it was handling things like that is what helped build up the confidence to talk to people, especially adults… (#6, non-Hispanic/Latina/o, White AYA, post-transfer).

Another participant described a similar shift of responsibility that started with mastering simple tasks before moving up to bigger responsibilities:

> I was a little older by then, and she (stepmom) was still washing my (nebulizer) mask. Even my social worker alluded to me having to do something simple that I should start doing it by myself just to, I guess, get ahead of the game because it's so simple. Then I had to start doing my appointments and all the insurance stuff, and it just became more natural to me. (#4, Hispanic/Latina/o AYA, pre-transfer).

In addition to discussing self-management of their physical health, participants also mentioned psychosocial issues such as peer relationships and mental health needs. For many participants, independent self-management extended beyond physical health and encompassed an increased awareness of their mental health. One participant described making this connection as they prepared to transition to adult care:

> I guess like having a chronic disease, obviously when you go to the pediatric (CF clinic) like it's a lot about like measuring things in your body, but there's a lot of things that are going on mentally. I think a big reason for depression is sometimes having a chronic disease. (#1, Hispanic/Latina/o AYA, post-transfer).

Preparing for change and the unknown

Preparing for change and the unknown encompasses the process of getting ready to move to the adult center, including concerns about leaving pediatrics, the experience of transition preparation and the process of redefining parent and child roles. When discussing the prospect of their impending transfer or reflecting upon their transfer experience, all participants shared that the adult CF center felt unknown to them, which for some participants caused fear or anxiety. “It's definitely going to be a bit nerve wracking. I think I'm pretty much okay with it. I mean what's different about it I'm not sure.” (#10, non-Hispanic/Latina/o, White AYA, pre-transfer) Often, this fear was related to leaving a team that had always cared for them.

> They (pediatric CF team) knew me there since I was a baby, so it was very – I was very comfortable going there. They seeing me [sic], saying hi, so it was like a regular thing. Every three months to seeing them, just extended family when I go there. (#5, non-Hispanic/Latina/o, Black AYA, post-transfer).

For others, early conversations with their pediatric CF team helped them to feel more prepared for adult care:

> I feel like the more time you have, the more prepared you will be. And that's why with me, when I was 16, that's when they started

mentioning and started talking to me about the life with CF as an adult and everything. (#12, Hispanic/Latina/o AYA, pre-transfer).

Few participants reported taking steps on their own to prepare themselves, such as learning about their illness and treatments:

I started to listen a lot more and do more research and start to learn more and learn about the reason of why I have to do the treatments and what the treatments could do to me and the medications that I consume. I just started to learn a lot more. (#9, Hispanic/Latina/o, Black AYA, pre-transfer).

Seven participants recalled being enrolled in CF R.I.S.E., a web-based transition preparation program used by this center. Of these, most had not interacted with it extensively if at all: "I think I had one meeting with someone and they sent me some website or something like that that lists everything but I didn't really look through it too much" (#10, non-Hispanic/Latina/o, White AYA, pre-transfer). For the few that did interact with the program, experiences varied. One participant expressed that they found the program unhelpful because it didn't reflect their personal CF experience and transition preparation needs:

…I took them all (CF RISE knowledge assessments) in an hour. I mean, like some of them I was just like, I was just clicking. I was like, this is the answer and you're telling me it's not right. Even though I know damn well it's right.
(#6, non-Hispanic/Latina/o, White AYA, post-transfer)

Another participant found the program to be a helpful way to learn about CF:

It did help me a lot with that, and that's what I liked about it, because I thought I knew what the CF was, but there was much more than what I knew. So, I learned a lot with those quizzes.
(#12, Hispanic/Latina/o AYA, pre-transfer)

Participants also offered suggestions to improve the transition preparation process for others in the future. One participant suggested that a conversation with their pediatric CF provider explaining how one would need to take on responsibility for speaking with doctors and making care decisions when one moved to adult care would have been helpful:

The other side of it, I think, is him saying or those pediatric doctors saying, "Listen, you will … it's a different experience over there. Your mom will not be the main leader of your team. You will be the leader of your team. You have to be ready for that." Even as that simple sentence, I think, would have helped prepare me a bit more.
(#7, non-Hispanic/Latina/o, White AYA, post-transfer)

Another participant echoed this sentiment by suggesting that in pediatrics, AYAs should have the opportunity to take on more responsibility for their care so that they feel prepared for the increased responsibility that is expected of them in the adult CF clinic: "I think that they should start patients at 16, 17, even the parents just doing...letting the patient do things on their own so they could be ready and there's no surprises to them once they hit the adult clinic." (#8, non-Hispanic/Latina/o, Black AYA, post-transfer). Other participants expressed a desire for more information regarding transition: "Maybe like even like little pamphlets it'd be like, okay this is what to expect." (#4, Hispanic/Latina/o AYA, pre-transfer) Another participant suggested that having a directory of resources and phone numbers would help them navigate the adult care environment:

I don't feel there was enough support systems posted on that like, do you know who you can contact in the CF center or the foundation, or even in the hospital, right? Numbers, offering numbers, I don't want to say like a search system or like an area where you can just like an index, you know, I have an issue with billing, who do I contact, you contact X in the billing office. Okay. You know, have you ever managed on billing before? No, I haven't. Here's a good way of handling that.
(#6, non-Hispanic/Latina/o, White AYA, post-transfer)

Preparing for the changes associated with transition also included redefining parent-child roles within the family. One participant reported placing boundaries on their mother's involvement in their care so that they could become more independent prior to transfer: "That's when it started to get much more I guess serious for me because I told my mom she can't really do much anymore" (#3, non-Hispanic/Latina/o, White AYA, pre-transfer). Another participant reported that it was their parent who initiated the setting of boundaries on their involvement:

She tried to start falling back once I hit 18, but she was still there. She was still on top of things, but she was trying to let me do stuff on my own. But once I got to adult CF, it was over. It was definitely over.
(#8, non-Hispanic/Latina/o, Black AYA, post-transfer)

Transition experiences vary

For those who had completed their transfer to adult care, their experiences varied ranging from smooth and planned, to abrupt and unprepared. For one participant who experienced a sudden transfer, this move felt like an abandonment by the pediatric CF team. "But when they moved me, I was sad. I was mad. I was very upset. I didn't want to leave…they just gave me away" (#8, non-Hispanic/Latina/o, Black AYA, post-transfer). All participants who experienced an abrupt transfer to adult care described a unilateral decision to transfer to adult care, either on the part of the participant or their provider. For one participant, their provider made the decision to transfer:

They just called me, they just gave me appointment for the adult side. I didn't even go back to the pediatric side for my normal appointment and then they switched me and tell me [sic]. They just straight gave me an appointment at the adult side and I just had to go straight from there.
(#5, non-Hispanic/Latina/o, Black AYA, post-transfer)

Another participant reported initiating their own transfer when they felt their care would benefit from moving to the adult CF center: "I just called up the Adult Center and I made an appointment. I didn't even tell my Children's Center I wasn't coming back; I didn't want to do it." (#6, non-Hispanic/Latina/o, White AYA, post-transfer) Transition experiences were noted to vary by race, with non-Hispanic/ Latina/o Black AYAs more frequently reporting a sudden transfer of care initiated by their provider. In contrast, participants who experienced a smooth and prepared transition often expressed being involved in decision making with their provider about the timing and planning of their transition.

They really didn't make, or put any pressure on me, because I was still young as well and it was just more like to start thinking about how I was going to be when I was older and moving on to this other part of my life and stuff with the CF. So, everything with my doctors went well. They never really pressured me, so that's why like I said that it took time, and I took my time with everything.
(#12, Hispanic/Latina/o AYA, pre-transfer)

Participants who experienced a smooth transition also often described experiencing a "warm handoff" between the pediatric and adult CF teams, a gesture that was received positively by participants who experienced it: "They like went with me to the adult section and just like introduced me and everything, so that was nice." (#1, Hispanic/Latina/o AYA, post-transfer).

Despite varying experiences of the transition process, all participants who had already transferred to adult care reported a positive experience once they arrived. Many also described enjoying open communication with their adult CF care providers. In particular, some participants noted that they felt more comfortable discussing adult health issues, such as reproductive health, with their adult CF provider. One participant contrasted these discussions in the pediatric and adult clinics:

> Even when I asked him (pediatric doctor) … I would ask him questions about reproductive health with CF, which is obviously a very important topic. He was just very squirmy about it. Another reason I think is I just feel very comfortable with (adult doctor) asking her anything. Anything, really, and she is very open to that. (#7, non-Hispanic/Latina/o, White AYA, post-transfer).

Another participant shared a similar experience of being able to openly communicate with their adult provider:

> When you move to the adult clinics, it's, you have more like, you just feel more comfortable to start talking about, adult things or more personal stuff. Because you're on the adult side, and they're like, they kind of reiterate that a lot too. You know, you're an adult. (#2, non-Hispanic/Latina/o, White AYA, post-transfer).

Discussion

This study updates and adds to the limited qualitative literature on transition experiences in CF (Al-Yateem, 2012; Brumfield & Lansbury, 2004), which was conducted prior to widespread availability of CFTR modulator therapies. The three themes identified in this study conceptualize the healthcare transition experience for a racially and ethnically diverse sample of AYAs with CF. The theme of *independent care of the whole self* represents the process of assuming CF self-management responsibility and the associated challenges and successes. *Preparing for change and the unknown* addresses the process of getting ready for adult care and coping with the associated uncertainty. Experiences of transition preparation interventions at this center were included in this theme, as well as how AYAs navigated changing parent and child roles during the transition preparation process. The final theme, *transition experiences vary*, describes how some AYAs experienced a smooth and planned transition characterized by shared decision making, while others experienced an abrupt transfer of care characterized by a unilateral decision to transfer either on the part of the AYA or their provider. This theme was the only one which appeared to diverge by race; Non-Hispanic/ Latina/o Black AYAs who had completed their transfer to adult care described a sudden and unprepared transfer of care which was initiated by their provider.

The theme, *independent care of the whole self,* represents the process of assuming CF self-management responsibility and the associated challenges and successes. An early and gradual shift of self-management responsibility from parent to child was cited by participants as something which helped them prepare for adult care. The importance of gradually transferring self-management responsibility beginning in adolescence has been reported in other studies of AYAs with CF and their families (Dupuis et al., 2010; Sawicki et al., 2015). The process of shifting self-management responsibility is common across chronic illnesses. For example, AYAs with chronic kidney disease and their parents described the benefits of incrementally increasing disease management responsibility beginning in mid-adolescence (Nicholas et al., 2018). Parents of AYAs with type 1 diabetes also described preferring a gradual process of handing over diabetes management to their child (Pritlove et al., 2020).

Participants shared mixed feelings about the transition preparation available to them at their CF center and offered constructive suggestions for improving the healthcare transition process. AYAs in our study desired adequate time to prepare and wanted additional information about adult care to ease their fear of the unknown. This is similar to the identified need for improved healthcare transition preparation and enhanced information about adult care reported in other chronic illnesses, including sickle cell disease (Clayton-Jones et al., 2021), juvenile idiopathic arthritis (Raunsbæk Knudsen et al., 2018) and cerebral palsy (Larivère-Bastien et al., 2013). Others have also noted that it is important to provide AYAs with information about the transition process and offer a chance to tour the adult hospital prior to transfer in CF (Brumfield & Lansbury, 2004) and type I diabetes (Garvey et al., 2014). Our findings support the development of transition preparation interventions which begin early in adolescence and provide AYAs with information about adult care and a sense of control over their transition. Participants expressed mixed feelings about the transition preparation program CF R.I.S.E.; to our knowledge participant views of this program have not previously been explored in the literature.

The theme of *transition experiences vary*, encompasses differences in transition preparation and transfer as well as the experience of receiving care in the adult CF clinic. Striking differences in transition experiences were noted, ranging from smooth and planned, to sudden and abrupt. Abrupt transitions were noted to be characterized by a unilateral decision to transfer, initiated by either the provider or the patient. This has been reported by others; for example, in a qualitative study of providers at a human immunodeficiency virus (HIV) clinic, three common transition experiences were identified: ideal, abrupt and de-facto (Hussen et al., 2019). However, to our knowledge, abrupt transitions have not previously been described in the limited qualitative literature on transition in CF. The descriptions of ideal and abrupt transitions align with our findings of smooth and sudden transition experiences. De-facto transition, or transfer to adult care when a patient reaches a cutoff age regardless of preparation, was not described by our participants as the pediatric clinic site for this study does not have a strict age limit for transfer (Hussen et al., 2019). To help close this gap in transition experiences, we recommend involving AYAs in decision making about their transition, which would help align decisions to transfer care, thus promoting smooth and planned care transitions.

Perhaps most importantly, our racially and ethnically diverse sample reported differences in transfer experience with Black AYAs with CF more frequently reporting that a provider made a unilateral decision to transfer care. Similar findings have been reported in studies of transition experiences in sickle cell disease (Bemrich-Stolz et al., 2015; Mulchan et al., 2016), a condition which disproportionately affects individuals of Black race, supporting a possible influence of race on abrupt healthcare transition experiences. A study of AYAs with sickle cell disease where 93% of participants identified as Black and healthcare providers also found that the AYA participants often experienced abrupt transfers to adult care combined with transition experiences which did not align with their expectations (Mulchan et al., 2016). Participants in the Mulchan et al. study (2016) often reported experiencing abrupt transfers to adult care and transition experiences which did not align with their expectations. Further, participants in the Bemrich-Stolz et al. study (2015) expressed feeling abandoned during their transition and described being first told of their impending transfer very shortly before it occurred (Bemrich-Stolz et al., 2015). Whether these experiences represent faulty recall on the part of the participant, implicit bias, punitive behavior, or discrimination on the part of the provider, or simply the absence of shared decision-making between the patient and the provider warrants further exploration.

It is also important to note that a transition preparation process had been recently established at this pediatric CF center in 2015, and the process may have been rushed in older AYAs who had been waiting to transfer. The transition process may also be different for those who start earlier in adolescence. It is likely that those who start the transition process earlier may have a smoother transition since they have longer to prepare. Another factor that may contribute to the varying transition experiences among patients is that each provider may approach transition preparation and transfer of care in different way. Therefore, we

recommend that the transition process should be standardized among providers at a given CF center.

Practice Implications

Pediatric nurses play an important role in the transition preparation process; results of this qualitative study have implications for pediatric nursing practice. Healthcare transition preparation should start early in adolescence with a gradual transfer of responsibility from parent to child. Nurses can support this transfer of responsibility by engaging adolescents as well as parents during care encounters and by offering support, education, and encouragement for independent self-management. We also recommend that pediatric nurses involve adolescents with chronic illness in planning for their transition and offer them a sense of control over the process which may support a smooth transition and decrease feelings of uncertainty.

Limitations

This study has several limitations. While qualitative research does not yield findings that are generalizable to a larger population, it is important to consider that the findings of this study reflect the experiences of a small sample of AYAs with CF from one medical center and their experiences are reflective of the transition preparation programming available at this center. Our sample is also limited in that female participants accounted for only 25% of our sample, whereas CF affects males and females approximately equally. The decision to limit attribution of quotes to only to characteristics most relevant to the context of the study was made to protect the privacy of participants. Additionally, we recognize that healthcare transition affects the entire family as well as healthcare providers. A limitation of this study is its representation of only the AYA perspective of healthcare transition; we recommend that future work include parent and healthcare provider perspectives. Conducting the interviews only in English may also be a limitation, however, we were able to recruit a representative proportion of Hispanic/Latina/o AYAs and only one AYA was ineligible due to an inability to obtain parent consent in English. We also recognize that recall bias affects our results, particularly for AYAs who had transitioned less recently. Additionally, social desirability bias may have affected the results, particularly regarding interview questions about self-management, likely resulting in an overreporting of self-management responsibility. Similarly, in responding to questionnaire items relating to self-management responsibility, there may have been a disconnect between reported self-management responsibility and actual task completion, which was not captured by the questionnaire. Finally, participants who were receiving regular CF care or who had notably positive or negative transition experiences may have been more likely to participate.

Conclusions

Understanding how AYAs with CF experience healthcare transition and transition preparation programs is key to improving healthcare transition experiences and outcomes for this population. Our findings suggest that not all AYAs experience transition the same way; practice changes to ensure that all AYAs receive tailored transition preparation that meets their individual needs are recommended to help close this gap in transition experiences. Ensuring that transition discussions start early, that transition interventions are informed by AYA and caregiver needs and preferences, and involve principles of shared decision-making, are initial steps that can be implemented to improve transition experiences. Further recommendations to improve transition preparation include promoting independent self-management prior to transfer of care and providing transition resources and information about adult care early in the transition preparation process. While understanding the AYA experience is integral to improving healthcare transition preparation, future research should focus on comprehensively evaluating healthcare transition programs in CF including exploring the parent perspective of transition preparation and evaluating other transition outcomes including lung function and adult CF clinic attendance, with a focus on the transition outcomes and experiences of racially and ethnically diverse AYAs with CF.

Funding source declaration

This work was supported by a research grant awarded by the Alpha Zeta Chapter of Sigma Theta Tau International and the National Institute of Nursing Research [1F31NR020141–01]. The funding sources had no role in study design or data collection, interpretation, or analysis.

CRediT authorship contribution statement

Katherine South: Conceptualization, Methodology, Investigation, Data curation, Formal analysis, Writing – original draft, Funding acquisition. **Maureen George:** Conceptualization, Formal analysis, Writing – review & editing, Supervision. **Hossein Sadeghi:** Supervision, Resources, Writing – review & editing. **Victoria Piane:** Resources, Writing – review & editing. **Arlene Smaldone:** Conceptualization, Formal analysis, Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

We would like to thank Dr. Emily DiMango and Dr. Claire Keating for their support of this study and their assistance in participant recruitment.

References

Ahern, K. J. (1999). Pearls, pith, and provocation: Ten tips for reflexive bracketing. *Qualitative Health Research*, *9*, Issue 3.

Al-Yateem, N. (2012). Child to adult: Transitional care for young adults with cystic fibrosis. *British Journal of Nursing*, *21*(14).

Baker, A. M., Riekert, K. A., Sawicki, G. S., & Eakin, M. N. (2015). CF RISE: Implementing a clinic-based transition program. *Pediatric Allergy, Immunology and Pulmonology*, *28*(4), 250–254. https://doi.org/10.1089/ped.2015.0594.

Bemrich-Stolz, C., Halanych, J., Howard, T., Hilliard, L., & Lebensburger, J. (2015). Adult patients with sickle cell disease. *Int J Hematol Ther*, 1–13.

Betz, C., Coyne, I., & Hudson, S. (2021). Health care transition: The struggle to define itself. *Comprehensive Child and Adolescent Nursing*, *39*(1), 1–15. https://doi.org/10.1080/24694193.2021.1933264.

Birt, L., Scott, S., Cavers, D., Campbell, C., & Walter, F. (2016). Member checking: A tool to enhance trustworthiness or merely a nod to validation? *Qualitative Health Research*, *26*(13), 1802–1811. https://doi.org/10.1177/1049732316654870.

Bradshaw, C., Atkinson, S., & Doody, O. (2017). Employing a qualitative description approach in health care research. *Global Qualitative Nursing Research*, *4*. https://doi.org/10.1177/2333393617742282.

Brumfield, K., & Lansbury, G. (2004). Experiences of adolescents with cystic fibrosis during their transition from paediatric to adult health care: A qualitative study of young Australian adults. *Disability and Rehabilitation*, *26*(4), 223–234. https://doi.org/10.1080/09638280310001644924.

Clayton-Jones, D., Matthie, N., Treadwell, M., Field, J. J., Mager, A., Sawdy, R., ... Haglund, K. (2021). Social and psychological factors associated with health care transition for young adults living with sickle cell disease. *Journal of Transcultural Nursing*, *32*(1), 21–29. https://doi.org/10.1177/1043659619896837.

Coyne, I., Sheehan, A. M., Heery, E., & While, A. E. (2017). Improving transition to adult healthcare for young people with cystic fibrosis: A systematic review. *Journal of Child Health Care*, *21*(3), 312–330. https://doi.org/10.1177/1367493517712479.

Cystic Fibrosis Education & Support Resource: CF R.I.S.E (2021). Retrieved July 17, 2020, fromhttps://www.cfrise.com/.

Cystic Fibrosis Foundation (2019). *2018 Patient Registry Annual Data Report.*

Davidson, L. F., & Doyle, M. H. (2021). Health-care transition: A vital part of care, growth, and change for pediatric patients. *Pediatrics in Review*, *42*(12), 684–693. https://doi.org/10.1542/pir.2020-000422.

Dupuis, F., Duhamel, F., & Gendron, S. (2010). Confidence, supervision and taking responsibility: Changes in the development of autonomy in families who live with an

adolescent with cystic fibrosis. *Enfermería Clínica*, *20*(2), 105–108. https://doi.org/10.1016/j.enfcli.2009.11.006.

Elo, S., & Kyngäs, H. (2008). The qualitative content analysis process. *Journal of Advanced Nursing*, *62*(1), 107–115. https://doi.org/10.1111/j.1365-2648.2007.04569.x.

Garvey, K. C., Beste, M., Luff, D., Atakov-Castillo, A., Wolpert, H., & Ritholz, M. (2014). Experiences of health care transition voiced by young adults with type 1 diabetes: A qualitative study. *Adolescent Health, Medicine and Therapeutics*, *5*, 191. https://doi.org/10.2147/ahmt.s67943.

Goralski, J. L., Nasr, S. Z., & Uluer, A. (2017). Overcoming barriers to a successful transition from pediatric to adult care. In Pediatric Pulmonology. *John Wiley and Sons Inc.*, *52* (S48), S52–S60. https://doi.org/10.1002/ppul.23778.

Guba, E. (1981). ERIC/ECTJ Annual review paper criteria for assessing the trustworthiness of naturalistic inquiries. *ECTJ*, *29*(2), 75–91.

Guest, G., Bunce, A., & Johnson, L. (2006). How many interviews are enough?: An experiment with data saturation and variability. *Field Methods*, *18*(1), 59–82. https://doi.org/10.1177/1525822X05279903.

Hsieh, H. -F., & Shannon, S. E. (2005). Three approaches to qualitative content analysis. *Qualitative Health Research*, *9*, 1277–1288. https://doi.org/10.1177/1049732305276687.

Hussen, S. A., Chakraborty, R., Camacho-Gonzalez, A., Njiemoun, B., Grossniklaus, E., Goodstein, E., ... del Rio, C. (2019). Beyond "purposeful and planned": Varied trajectories of healthcare transition from pediatric to adult-oriented care among youth living with HIV. *AIDS Care - Psychological and Socio-Medical Aspects of AIDS/HIV*, *31*(1), 45–47. https://doi.org/10.1080/09540121.2018.1488029.

Johnson, C., Butler, S. M., Konstan, M. W., Morgan, W., Ellen, M., & Wohl, B. (2003). Factors influencing outcomes in cystic fibrosis* a center-based analysis. *Chest*, *123*(1). www.chestjournal.org.

Kallio, H., Pietilä, A. M., Johnson, M., & Kangasniemi, M. (2016). Systematic methodological review: Developing a framework for a qualitative semi-structured interview guide. *Journal of Advanced Nursing*, *72*(12), 2954–2965 Blackwell Publishing Ltdhttps://doi.org/10.1111/jan.13031.

Kerr, C., Nixon, A., & Wild, D. (2010). Assessing and demonstrating data saturation in qualitative inquiry supporting patientreported outcomes research. *Expert Review of Pharmacoeconomics & Outcomes Research*, *10*(3), 269–281. https://doi.org/10.1586/erp.10.30.

Lanzkron, S., Sawicki, G. S., Hassell, K. L., Konstan, M. W., Liem, R. I., & McColley, S. A. (2018). Transition to adulthood and adult health care for patients with sickle cell disease or cystic fibrosis: Current practices and research priorities. *Journal of Clinical and Translational Science*, *2*(5), 334–342. https://doi.org/10.1017/cts.2018.338.

Larivère-Bastien, D., Bell, E., Majnemer, A., Shevell, M., & Racine, E. (2013). Perspectives of young adults with cerebral palsy on transitioning from pediatric to adult healthcare systems. *Seminars in Pediatric Neurology*, *20*(2 PG-154–159), 154–159. https://doi.org/10.1016/j.spen.2013.06.009.

MacQueen, K. M., McLellan, E., Kay, K., & Milstein, B. (1998). Codebook development for team-based qualitative analysis. *CAM Journal*, *10*(2), 31–36. https://doi.org/10.1177/1525822X980100020301.

McGarry, M. E., & McColley, S. A. (2021). Cystic fibrosis patients of minority race and ethnicity less likely eligible for CFTR modulators based on CFTR genotype. *Pediatric Pulmonology*, *56*(6), 1–8 December 2020https://doi.org/10.1002/ppul.25285.

McGarry, M. E., Williams, W. E. I., & Mccolley, S. A. (2019). The demographics of adverse outcomes in cystic fibrosis. *Pediatric Pulmonology*, *54*. https://doi.org/10.1002/ppul.24434.

Middleton, P. G., & Taylor-Cousar, J. L. (2021). Development of elexacaftor–tezacaftor–ivacaftor: Highly effective CFTR modulation for the majority of people with cystic fibrosis. *Expert Review of Respiratory Medicine*, *15*(6), 1–13. https://doi.org/10.1080/17476348.2021.1855980.

Mulchan, S., Valenzuela, J., Crosby, L., Sang, D. P., & C. (2016). Applicability of the SMART model of transition readiness for sickle-cell disease. *Journal of Pediatric Psychology*, *41* (5), 543–554. https://doi.org/10.1093/jpepsy/jsv120.

Nicholas, D. B., Kaufman, M., Pinsk, M., Samuel, S., Hamiwka, L., & Molzahn, A. E. (2018). Examining the transition from child to adult Care in Chronic Kidney Disease: An open exploratory approach. *Nephrology Nursing Journal : Journal of the American Nephrology Nurses' Association*, *45*(6), 553–559. https://ezproxy.cul.columbia.edu/login?qurl=https%3A%2F%2Fsearch.ebscohost.com%2Flogin.aspx%3Fdirect%3Dtrue%26db%3Dcin20%26AN%3D133734034%26site%3Dehost-live%26scope%3Dsite.

Okumura, M. J., & Kleinhenz, M. E. (2016). Cystic fibrosis transitions of care lessons learned and future directions for cystic fibrosis. *Clinics in Chest Medicine*, *37*. https://doi.org/10.1016/j.ccm.2015.11.007.

Padman, R., McColley, S. A., Miller, D. P., Konstan, M. W., Morgan, W. J., Schechter, M. S., ... Wagener, J. S. (2007). Infant care patterns at epidemiologic study of cystic fibrosis sites that achieve superior childhood lung function. *Pediatrics*, *119*(3), e531–e537. https://doi.org/10.1542/peds.2006-1414.

Patton, M. Q. (1990). *Qualitative evaluation and research methods*. SAGE Publications, inc.

Pierce, J. S., Aroian, K., Milkes, A., Schifano, E., Schwindt, T., Gannon, A., ... Wysocki, T. (2017). Health care transition for young adults with type 1 diabetes: Stakeholder engagement for defining optimal outcomes. *Journal of Pediatric Psychology*, *42*(9), 970–982. https://doi.org/10.1093/jpepsy/jsx076.

Pritlove, C., Markowitz, B., Mukerji, G., Advani, A., & JA, P. (2020). Experiences and perspectives of the parents of emerging adults living with type 1 diabetes. *BMJ Open Diabetes Research & Care*, *8*, 1. https://doi.org/10.1136/bmjdrc-2019-001125.

Raunsbæk Knudsen, L., de Thurah, A., & Bjerrum, M. (2018). Transition from child to adult care in an outpatient clinic for adolescents with juvenile idiopathic arthritis: An inductive qualitative study. *Nursing Open*, *5*(4), 546–554. https://doi.org/10.1002/nop2.164.

Rosen, D., Blum, R. W., Britto, M. T., Sawyer, S. M., & Siegal, D. (2003). Transition to adult health Care for Adolescents and Young Adults with Chronic Conditions. *Journal of Adolescent Health*, *03*, 309–311. https://doi.org/10.1016/S1054-139X(03)00208-8.

Sandelowski, M. (2000). Whatever happened to qualitative description? *Research in Nursing & Health*, *23*(4), 334–340 http://doi.org/10.1002/1098-240x(200008)23:4<334::aid-nur9>3.0.co;2-g.

Sawicki, G. S., Heller, K. S., Demars, N., & Robinson, W. M. (2015). Motivating adherence among adolescents with cystic fibrosis: Youth and parent perspectives. *Pediatric Pulmonology*, *50*(2), 127–136. https://doi.org/10.1002/ppul.23017.

Sawicki, G. S., Ostrenga, J., Petren, K., Fink, A. K., D'Agostino, E., Strassle, C., ... Rosenfeld, M. (2018). Risk factors for gaps in care during transfer from pediatric to adult cystic fibrosis programs in the United States. *Annals of the American Thoracic Society*, *15*(2), 234–240. https://doi.org/10.1513/AnnalsATS.201705-357OC.

Society for Adolescent Health and Medicine (2020). Transition to adulthood for youth with chronic conditions and special health care needs. *Journal of Adolescent Health*, *66*(5), 631–634. https://doi.org/10.1016/j.jadohealth.2020.02.006.

Sullivan-Bolyai, S., Bova, C., & Harper, D. (2005). Developing and refining interventions in persons with health disparities: The use of qualitative description. *Nursing Outlook*, *53*, 127–160. https://doi.org/10.1016/j.outlook.2005.03.005.

White, P. H., & Cooley, W. C. (2018). Supporting the health care transition from adolescence to adulthood in the medical home. *Pediatrics*, *142*(5). www.aappublications.org/news.